CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – III

HARDENED CONCRETE

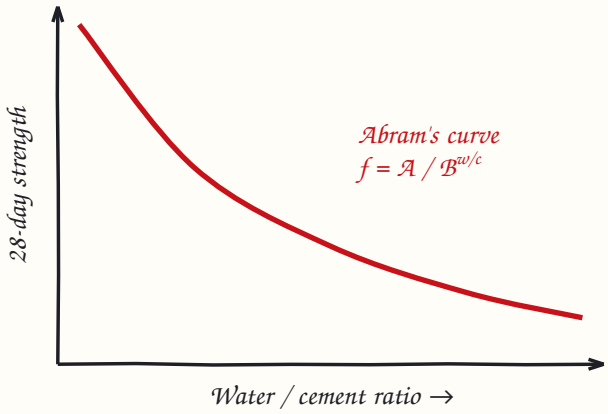
Syllabus covered in this unit

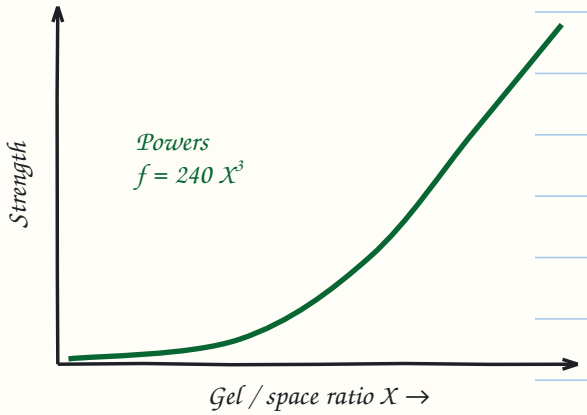
Water/cement ratio – Abram's law – gel/space ratio – nature of strength – maturity concept – strength in tension & compression – factors affecting strength – relation between compressive & tensile strength – curing – testing of hardened concrete : compression, tension, flexure, splitting – non-destructive testing – codal provisions

1. WATER / CEMENT RATIO

- **w/c ratio** = weight of water / weight of cement in the mix. It is the **single most important factor controlling strength, permeability and durability** of concrete.
- Water needed for hydration ≈ 0.23 ; for filling gel pores ≈ 0.15 → **0.38** for complete hydration. Excess water leaves **capillary pores** after evaporation → porosity ↑, strength ↓.
- If w/c < 0.38 : full hydration impossible (unhydrated cement acts as filler) but still strength high with SP; for w/c > 0.38 : capillary pores remain.
- Practical range : 0.25 (HPC) – 0.65 ; **IS 456 limits max w/c by exposure** :

Exposure	Min. grade RCC	Min. cement (kg/m^3) RCC	Max w/c (RCC)
Mild	M20	300	0.55
Moderate	M25	300	0.50
Severe	M30	320	0.45
Very severe	M35	340	0.45
Extreme	M40	360	0.40





1.1 Abram's law (D.A. Abrams, 1918)

Abram's water–cement ratio law

"For given materials and workable mix, the strength of fully compacted concrete at a given age is **inversely proportional to the w/c ratio**."

$f_c = A / B^{\chi}$ (χ = w/c by volume) Abram : $f_c = 14\,000 / 7^{\chi}$ psi

$A \approx 96$ MPa, $B \approx 4$ (SI, by weight) – A, B depend on cement, age, curing, aggregate.

- Plot : f_c vs w/c is a **hyperbola** ; **log f_c vs w/c is a straight line** ($\log f = \log A - \chi \log B$).
- **Valid** for workable, **fully compacted** mixes, same materials & curing ; fails for very low w/c (poor compaction) and for very high cement contents.
- Other relations : **Feret's** $f = K[c / (c + w + a)]^2$ (c, w, a = absolute volumes of cement, water, air) ; **Bolomey** $f = K(c/w - 0.5)$.
- **Limitation** : not consider air voids, quality of aggregate, age, etc. – Feret's is better.

2. GEL / SPACE RATIO (POWERS)

- **Gel/space ratio** X = volume of hydrated cement paste (gel) / (volume of gel + capillary pores) = **solid products ÷ available space.**

Powers' formula

$X = 0.657\,\alpha / (0.319\,\alpha + w/c)$ Strength $f_c = 240\,X^3$ (N/mm^2)

α = degree of hydration (0 – 1). Gel volume = 2.06 × vol. of cement hydrated. (Max gel/space ratio = 1.0 → 240 MPa theoretical)

- Example : w/c = 0.5, $\alpha = 0.8 \rightarrow X = 0.657 \times 0.8 / (0.319 \times 0.8 + 0.5) = 0.5256 / 0.7552 =$ **0.696** ; $f = 240 \times 0.696^3 =$ **81 MPa** (lab-paste, ≈ gel-space theory gives higher than 28-d concrete, which also depends on aggregate, ITZ, etc.).

- Strength increases as **capillary porosity ↓ (w/c ↓, α ↑)**; w/c = 0.38 → X max (1.0) when α = 1.
- Powers' theory compares **different ages & w/c on a common basis**; (Abram's law considered only w/c).

3. NATURE OF STRENGTH OF CONCRETE

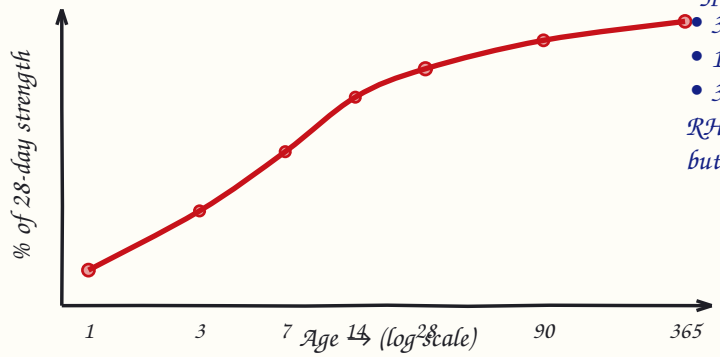
- Strength = capacity to **resist stress without failure** – compressive is the main strength ; tensile ≈ 8 – 12 % of it.
- Sources of strength : (i) **cohesion & adhesion** of hardened cement gel (**Van der Waals' forces**, chemical bonding) ; (ii) friction and **bond between paste and aggregate** (ITZ) ; (iii) mechanical interlock of aggregate.
- Concrete is a **three-phase composite** : aggregate – paste – **ITZ**; ITZ ($Ca(OH)_2$ rich, porous, micro-cracked) is the **weakest link** → micro-cracks exist even before loading (bond cracks).
- **Failure mechanism** (compression) : up to ~30 % of ultimate – stable, no new cracks ; 30–50 % – bond cracks grow ; 50–75 % – **mortar cracks** start & join ; 75–100 % – unstable crack propagation → **hourglass / cone-shear failure** (lateral tensile strain).
- Strength of the paste & aggregate > concrete strength → **ITZ governs** ; hence silica fume (densifies ITZ) ↑ strength.

4. FACTORS AFFECTING STRENGTH OF CONCRETE

1. **w/c ratio** : lower → higher strength (Abram).
2. **Cement** : type (RHC, OPC 53), fineness, compounds (C_3S, C_2S), **cement content** (strength ↑ with richness upto optimum).
3. **Aggregates** : strength, **maximum size** (small better for high strength), shape (angular ↑ flexural), **texture** (rough ↑ bond), **grading** (↑ density), cleanliness.
4. **Degree of compaction** : 5 % voids → ~30 % strength loss.
5. **Curing** : moisture + temperature ; 7-day cured gets ~ 90 % of 28-day wet-cured (OPC) ; air-cured loses about 40 %.
6. **Age** – strength increases with age if moisture available (see curve).
7. **Temperature** : high early temperature → faster strength gain but lower long-term ; below 5 °C slow ; freezing before 3.5 MPa disastrous.
8. **Admixtures & pozzolans** : plasticizers ↑ (lower w/c) ; fly ash ↓ early ↑ later ; silica fume

↑↑ ; air entrainment ↓ 5 % per 1 % air ; $CaCl_2$ ↑ early.

9. **Mixing water quality ; mixing time & method ; rate of loading** (faster → higher test strength) ; **specimen size, shape & moisture condition** (wet specimen weaker 10–15 %).
10. **Size effect** : larger specimen gives lower strength (cylinder 150×300 ≈ 0.8 × cube ; 100 mm cube ≈ 1.05 × 150 mm cube).



Typical OPC gain with age (moist cured)
• 3 d ≈ 40 % • 7 d ≈ 65 %
• 14 d ≈ 90 % • 28 d = 100 % (reference)
• 3 months ≈ 110 % • 1 year ≈ 120 %
RHC gains faster; PPC / slag slower initially but reaches higher ultimate strength

5. MATURITY CONCEPT

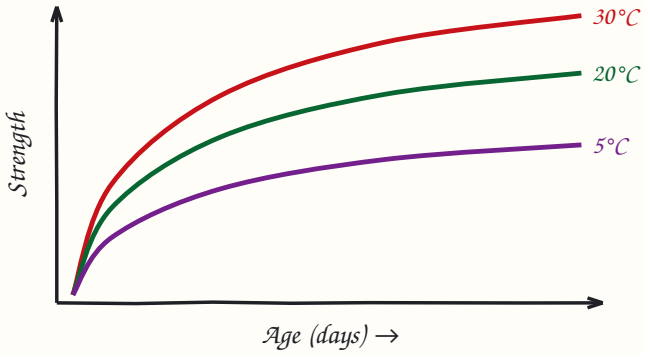
- **Maturity** = the extent of strength development as a function of **age and curing temperature** ; concrete of the same maturity has the same strength regardless of its temperature–time history (for same mix).

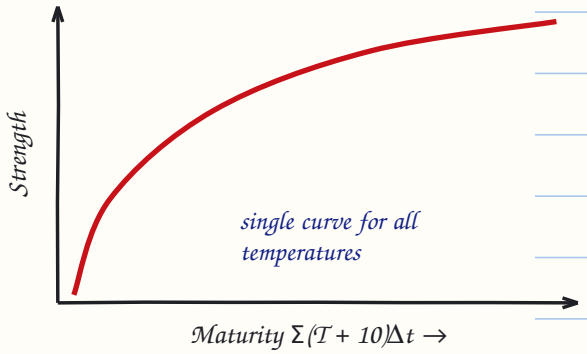
Nurse–Saul maturity function

$M = \Sigma\ (T - T_0)\,\Delta t$ (°C·h or °C·days) ; T_0 = datum temperature = −10 °C (some codes −11 °C or 0 °C)

Strength $S = a + b \cdot \log_{10}(M)$ (Plowman) ; Equivalent age $t_e = \Sigma\ (T - T_0)\,\Delta t\ /\ (T_r - T_0)$

- Example : concrete cured at 20 °C for 28 days : $M = (20 + 10) \times 28 \times 24$ = **20 160 °C·h** – concrete at 10 °C reaches same maturity in 20 160/(20×24) = 42 days.
- Uses : estimating strength **in-situ** from temperature records (thermocouples), deciding **de-shuttering / prestressing** time, cold weather concreting, steam-cured precast, mass concrete.
- Limitations : ignores initial curing temperature effects, moisture; strength–maturity curve must be calibrated for each mix; accuracy ±10–15 %.





6. STRENGTH IN COMPRESSION

- **Characteristic strength** (f_{ck}) = strength below which not more than **5 %** of test results are expected to fall = f_{mean} **– 1.65 s**.
- Grades (IS 456) : **M10, M15, M20, M25, M30, M35, M40 … M80** (M = mix, number = f_{ck} in MPa at 28 days on 150 mm cubes). Ordinary M10–M20; standard M25–M55; high strength M60–M80.
- **Design strength** $f_d = f_{ck} / \gamma_m = 0.67\ f_{ck} / 1.5 =$ **0.446** f_{ck} (limit state).
- Stress–strain curve of concrete: parabolic upto 0.002 strain then flat to $\varepsilon_{cu} =$ **0.0035** (IS 456).
- **Factors** : w/c, cement, aggregate, age, compaction, curing (see section 4).

7. STRENGTH IN TENSION

- Direct tension is difficult to apply (grip problems, eccentricity). Hence 3 indirect tests :
  - **Flexural (modulus of rupture)** $f_{cr} = 0.7\sqrt{f_{ck}}$ (IS 456 cl. 6.2.2)
  - **Split tensile** $\approx 0.56 – 0.66\sqrt{f_{ck}}$ (ACI : $0.56\sqrt{f'c}$) – slightly higher than direct
  - **Direct tension** $\approx 0.33\sqrt{f'_c}$ (ACI) – lowest.
- Why low tensile strength : micro-cracks at ITZ propagate rapidly under tension (brittle).
- Importance : **cracking load** of beams, slabs on grade, pavements, prestressed concrete (cracking moment), shear & bond.

Relation between compressive and tensile strength

Type	Formula (N/mm^2)	For M30
Flexural f_{cr} (IS 456)	$0.7\sqrt{f_{ck}}$	3.83
Split tensile f_{sp}	$\approx 0.5 – 0.6\sqrt{f_{ck}}$	2.7 – 3.3

Type	Formula (N/mm^2)	For M30
Direct tension f_t	$\approx 0.35 – 0.5\sqrt{f_{ck}}$	1.9 – 2.7
Order	$f_c \gg f_{cr} > f_{sp} > f_t$	$f_{cr} / f_{ck} \approx$ 10–15 %

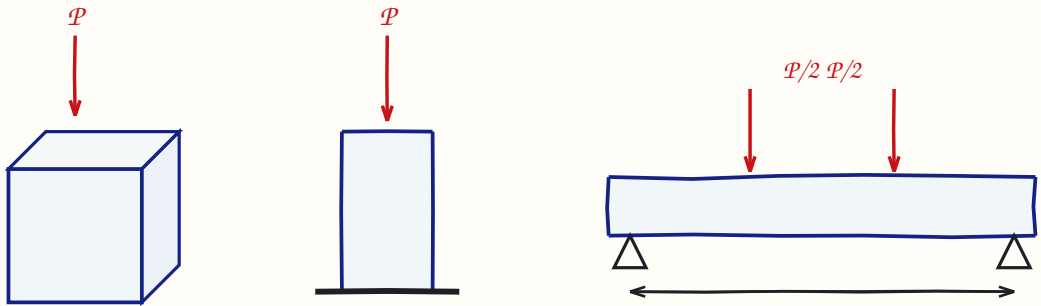
- Ratio decreases as compressive strength increases (a non-linear relation $f_t \propto f_c^{2/3}$). Tensile strength more sensitive to **curing, w/c, aggregate texture**.

8. CURING (EFFECT ON STRENGTH)

- Curing = keeping concrete moist & warm after placement to ensure hydration ; **continuous curing for 28 days gives max strength**.
- **Without curing** the strength at 28 d may be ≈ 50 % of that of properly cured concrete; 3 days wet curing → 60–70 %; 7 days → 90 %; 14 days → 95 %.
- Curing also reduces shrinkage cracking, permeability & improves **durability, abrasion resistance**.
- **Curing methods** (ponding, spraying, membrane, steam) are covered in Unit II.
- Cold weather curing > 5 °C ; hot weather – prevent evaporation (> 1 $kg/m^2/h$ → plastic shrinkage cracks).

9. TESTING OF HARDENED CONCRETE

9.1 Compressive strength test (IS 516 : 1959)

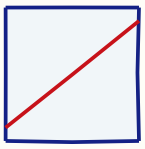


- **Specimen** : cube **150 × 150 × 150 mm** (100 mm if MSA ≤ 20 mm) or cylinder **150 dia × 300 mm** (IS 516) ; mould of cast iron / steel, oiled.
- **Casting** : fill in **3 layers** (≈ 50 mm each), **35 strokes per layer** with 25 mm square steel bar (or vibrate on table) ; level, cover with plate ; **demould after 24 ± ½ h**.
- **Curing** : in clean water at **27 ± 2 °C** until test (**3, 7, 28 days**); tested in **saturated surface-dry** state.

- **Testing** : place cube **sideways** (cast faces not in contact) in CTM, **load at 14 N/mm^2/min (= 140 kg/cm^2/min = 0.25 N/mm^2/s)** till failure ; $f_c = P / A$.
- **Minimum 3 specimens** per sample ; mean = result ; if one value differs > ± 15 % from the median it is rejected (IS 516 : individual variation ≤ ± 15 % of mean).
- Cube vs cylinder : $f_{cyl} \approx 0.8\ f_{cube}$ (for normal-strength concrete).



Satisfactory (hour-glass)



Unsatisfactory (shear / side)

Beam failure position (a = distance of crack from nearer support, b = 150 mm)
- a ≥ 133 mm : $f_{cr} = P \cdot l / (b \cdot d^2)$
- 110 ≤ a < 133 mm : $f_{cr} = 3P \cdot a / (b \cdot d^2)$
- a < 110 mm : discard result

9.2 Flexure test (IS 516)

- Beam **150 × 150 × 700 mm** (100 × 100 × 500 if MSA < 20) ; cured in water, tested **wet**.
- **Two-point (third-point) loading**, span **600 mm** (for 150 mm beam) – pure bending in middle third → shear free zone ; load rate **0.7 N/mm^2/min** (400 kg/min for 150 beam).
- **Modulus of rupture** (if crack in middle third, a ≥ 133 mm) :

Flexural strength (third-point loading)

$f_{cr} = P \cdot l / (b \cdot d^2)$ (a > 133 mm) $f_{cr} = 3\ P \cdot a / (b \cdot d^2)$ (110 < a < 133 mm)

Centre-point loading : $f_{cr} = 3\ P\ l / (2\ b\ d^2)$ P = max load (N), l = span, b, d = width & depth

- Gives **higher value than direct tension** (1.5–2×) because of non-linear stress and stress gradient ; used for **pavement slabs** design.

9.3 Splitting tensile test (Brazilian test – IS 5816)

- Cylinder **150 mm dia × 300 mm long** laid horizontally between plates (hardboard / plywood strips 3–4 mm thick, 15 mm wide) in CTM ; compression applied along a **diameter** → a nearly uniform **horizontal tensile stress** develops across the vertical diameter plane and the cylinder splits along it.

Split tensile strength

$f_{sp} = 2\ P / (\pi\ D\ L)$ Rate of loading 1.2 – 2.4 N/mm^2/min (split tensile stress rate)

P = max load, D = diameter (150 mm), L = length (300 mm) → e.g. P = 200 kN → f_{sp} = $2 \times 200000 / (\pi \times 150 \times 300) = 2.83\ N/mm^2$

- Advantages : **simple, same cylinders as for compression, more uniform than flexure**, close to direct tension (5–12 % higher).

9.4 Direct tension – briefly

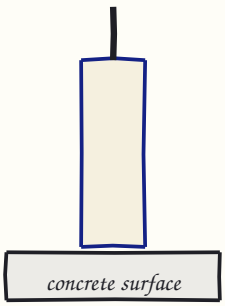
- Dog-bone / briquette specimens with embedded steel grips; hardly used because of alignment (eccentricity) problems.

10. NON-DESTRUCTIVE TESTING (NDT)

- NDT evaluates **strength, quality and defects** of in-situ concrete **without damage** ; used for old structures, cube failures, uniformity, crack depth, corrosion, repairs. **Core cutting** is semi-destructive (IS 516 Part 4).
- IS 456 cl. 17.9 : if cubes fail, NDT per **IS 13311 (Part 1 & 2)** & core test (**IS 516**); core strength ≥ 85 % of fck (avg) and no single core < 75 %.

10.1 Rebound (Schmidt) hammer & UPV

Schmidt rebound hammer



Rebound hammer (IS 13311 Pt 2) – surface hardness
- Spring-driven mass strikes plunger; rebound index R
- 9–15 readings per spot, reject outliers; R vs fck chart
- Orientation correction; affected by smoothness, moisture, carbonation, aggregate type, age
- Quick, cheap; accuracy ± 25 %

Ultrasonic Pulse Velocity (IS 13311 Pt 1)
- $V = L / t$; transducers 54 kHz, couplant grease
- >4.5 km/s excellent · 3.5–4.5 good · 3.0–3.5 doubtful · <3.0 poor

- UPV factors : moisture ↑ velocity, steel bars (↑), cracks/voids (↓), aggregate type, path length ≥ 100 mm (≥ 3 × MSA), temperature ; **Dynamic modulus** $E_d = \rho V^2 (1+\mu)(1-2\mu)/(1-\mu)$; transmission modes **direct, semi-direct, indirect (surface)**.
- **Combined method** (SONREB : rebound + UPV) – better correlation, reduces effect of moisture/age.

10.2 Other NDT methods

Method	Principle	Use
Penetration resistance (Windsor probe)	Steel probe fired by powder charge ; depth of penetration ∝ strength	Surface & near-surface strength
Pull-out test (Lok test)	Force needed to pull embedded steel insert with conical failure ≈ shear strength	Strength, de-shuttering time
Pull-off test	Tensile bond of overlays / repairs	Bond strength
Core test	100/150 mm dia core cut by diamond drill, ratio h/d = 2 (correction factors if 1–2), end-ground, tested saturated	Direct in-situ strength
Half-cell potential (ASTM C876)	$Cu/CuSO_4$ electrode vs steel : > −200 mV (90 % no corrosion) ; < −350 mV (90 % corrosion)	Reinforcement corrosion
Carbonation depth	Phenolphthalein spray : pink = uncarbonated (pH>9)	Depth of carbonation
Cover meter / GPR	Electromagnetic induction / radar	Rebar location, cover, spacing
Radiography , gamma rays	Absorption of X/γ-rays	Voids, honeycomb, steel detailing
Impact-echo / Acoustic emission / Infrared thermography	Stress waves, noise, heat patterns	Cracks, delaminations
Resistivity, chloride tests	Electrical resistivity, acid-soluble Cl content	Durability

11. CODAL PROVISIONS (IS 456)

- **Sampling frequency (cl. 15.2)** : 1–5 m^3 : 1 sample ; 6–15 : 2 ; 16–30 : 3 ; 31–50 : 4 ; > 50 : 4 + 1 per additional 50 m^3 or part thereof.
- **Test specimen** : 3 cubes per sample (7-day for early check, 28-day for acceptance).

Acceptance criteria – IS 456 cl. 16.1 (Table 11) ★

- **Mean of group of 4 non-overlapping consecutive results** ≥ f_{ck} + 0.825 s (s = established SD, min. as per Table 8) or f_{ck} + 3 N/mm^2 (M20 and above), whichever is greater.
- **Each individual result** ≥ f_{ck} − 3 N/mm^2 (M20 and above).
- If fails → NDT, core test ; **cores** : average ≥ 85 % f_{ck}, individual ≥ 75 % f_{ck} ; else structural load test (cl. 17.6) at 28 days + .

- **Assumed standard deviation (Table 8)** : M10, M15 – 3.5 ; M20, M25 – 4.0 ; M30–M50 – 5.0 N/mm^2.

- **Codes for NDT** : IS 13311 Pt 1 (UPV), Pt 2 (rebound hammer), IS 516 (Part 5 – NDT of hardened concrete), IS 456 cl. 17.9 ; ASTM C805 (rebound), C597 (UPV), C900 (pull-out), BS 1881 Part 201–207.

12. ACCELERATED STRENGTH TESTING (IS 9013)

- Purpose : predict 28-day strength within 24–28 hours for quality control & mix adjustment.
- **Warm water method** : cubes cured at 55 °C ± 2 for 20 h ; **boiling water method** : 3.5 h boiling after 23 h ; **autogenous** curing at 82 °C. Result correlated to 28-day strength by a pre-established line : $f_{28} = a + b \cdot f_{acc}$.
- Used in RMC plants for fast feedback, not for acceptance.

13. DURABILITY LINK AND PERMEABILITY

- Lower w/c → lower **permeability** (k falls from 10^{-9} to 10^{-13} m/s as w/c falls 0.7 → 0.4) → better resistance to sulphates, chlorides, carbonation, freeze–thaw, abrasion.
- Strength is usually a good index of durability **but** high strength concrete can still crack by thermal / shrinkage effects ; always check cover & curing.
- **Carbonation** : $CO_2 + Ca(OH)_2 \rightarrow CaCO_3$; pH falls from 12.5 to < 9 → steel passivity lost (depth ∝ $\sqrt{t}$).
- **Chloride limit** (IS 456 Table 7) : total acid-soluble Cl^- ≤ 0.6 kg/m³ for RCC ; sulphate (SO_3) ≤ 4 kg/m³.

14. SOLVED NUMERICALS

Problem 1 – cube strength

A 150 mm cube fails at **720 kN**. Find compressive strength. Cylinder strength?

Solution

$f_{cube} = P / A = 720 \times 10^3 / (150 \times 150) =$ **32.0 N/mm²** ; $f_{cyl} \approx 0.8 \times 32 =$ **25.6 N/mm²**

Problem 2 – Flexural strength

Beam 150×150×700, span 600 mm, third-point loading, fails at **12 kN** with crack at 200 mm from support.

Solution (a = 200 mm > 133 mm)

$f_{cr} = P\,l / (b\,d^2) = 12\,000 \times 600 / (150 \times 150^2) = 7.2 \times 10^6 / 3.375 \times 10^6 =$ **2.13 N/mm²**

Problem 3 – Using Abram's law

For a mix, A = 96, B = 4 : find f_c for w/c = 0.45 and 0.60.

Solution

$f_c = 96 / 4^{w/c}$: w/c = 0.45 → $4^{0.45} = 1.866$ → **51.4 MPa** ; w/c = 0.60 → $4^{0.60} = 2.297$ → **41.8 MPa**

Problem 4 – Maturity

Concrete cured at 15 °C for 10 days then 25 °C for 5 days. Find maturity ($T_0 = -10$ °C) and equivalent age at 20 °C.

Solution

$M = (15 + 10) \times 10 \times 24 + (25 + 10) \times 5 \times 24 = 6000 + 4200 =$ **10 200 °C·h**

Equivalent age at 20 °C = 10 200 / ((20 + 10) × 24) = **14.2 days**

Problem 5 – UPV

Path length 300 mm, transit time 66 µs. $V = 0.3 / 66\times10^{-6} =$ **4545 m/s = 4.55 km/s →** **excellent quality** concrete.

Problem 6 – Split tensile

Cylinder 150 × 300 mm fails at 180 kN : $f_{sp} = 2 \times 180\,000 / (\pi \times 150 \times 300) =$ **2.55 N/mm²**.

15. IMPORTANT QUESTIONS (EXAM)

1. State and explain Abram's law. What are its limitations?
2. Explain gel/space ratio & derive its relation with strength (Powers).
3. Discuss the factors affecting compressive strength of concrete.
4. Explain the maturity concept with an example. Where is it used?
5. Describe the procedure of the flexure test and split tensile test with formulae.
6. List NDT methods ; explain rebound hammer & UPV with their limitations.
7. Relationship between compressive strength and tensile strength of concrete.
8. IS 456 acceptance criteria for concrete cubes.

Quick revision – Unit III ★

- Abram $f = A/B^{w/c}$; Powers $f = 240\,X^3$, $X = 0.657\alpha/(0.319\alpha + w/c)$; maturity $M = \Sigma(T+10)\Delta t$.
- Cube 150 mm, 3 layers × 35, 27 ± 2 °C, load 14 N/mm²/min ; flexure $f_{cr} = Pl/bd^2$; split $f_{sp} = 2P/\pi DL$; $f_{cr} = 0.7\sqrt{f_{ck}}$.
- NDT : rebound hammer, UPV (>4.5 km/s excellent), pull-out, core, half-cell.